

Stat 301 Notes

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Course Goals for STAT 301

STAT 301 is an introduction to the theory underlying statistical inference. By the end of the course, you will:

- Understand different methods of obtaining point estimates, including maximum likelihood, method of moments, and Bayesian estimation
- Understand that some estimators are better than others, and be able to evaluate estimators
- Understand the theory underlying hypothesis tests, including errors, power, size, and level of tests
- Be able to derive hypothesis tests for simple situations from first principles, including likelihood ratio tests and most powerful tests
- Be able to derive interval estimates by inverting tests and using pivotal quantities

1 Introduction